UTS

Telecommunication Company: Marketing Campaign Performance

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Statistical Thinking for Data Science

Problem Formulation

1. Business Context

A telecommunications firm has initiated a marketing campaign aimed at introducing a new subscription offer to its clients. The objective of this analysis is to find out the attributes of the customers as well as the features of the campaign that have been effective in subscription. The data set consists of demographic, financial and other characteristics of the customers, who have been contacted as part of the campaign.

For the company it is helpful to identify the characteristics of customers who will subscribe to the offer. Nevertheless, the data set lacks the information about the features of the new offer and its price as well as the subscription plans of the customers prior to the campaign. Hence, the focus of the analysis is on observed relations between variables among the customers contacted during the campaign.

2. Analytical Problem

The aim here would be to establish the characteristics of the customers/campaigns that lead to subscriptions and which characteristics could be used in future modelling for customer targeting. These four dimensions include demographic traits, financial traits, previous experience in campaigns, and present campaign traits. Observational nature of the study and the absence of an untreated control group imply that identified relationships should be understood as correlations, not causalities. It is especially true for interaction-specific attributes like call duration since it cannot be used for customer targeting before contacting them.

3. Research Questions

The exploratory data analysis involves answering five questions:

1. Demographic features – Do the subscription rates vary among age, occupation, education, and marital status categories?
2. Financial and debt features – Are there any associations between financial and debt features and subscription rates?
3. Previous campaign participation – Is the previous campaign participation, including the frequency and outcome of contacts, related to subscription?
4. Current campaign contact – Are the contact method, timing, frequency, and duration related to subscription rates?
5. Customer segmentation – Are there any customer and campaign feature combinations that create customer segments with unusually high subscription rates?

4. Hypotheses

In view of the business setting and the variables at hand, the following exploratory hypotheses are formulated:

- H1: Subscription rates vary according to age, occupation, education, and marital status groups.
- H2 : Subscriptions vary based on the customers' financial and debt attributes.

- H3 : Better outcomes of past campaigns are related to higher subscription rates.
- H4 : Campaign interaction attributes, such as mode of interaction, time of interaction, frequency of interaction, and length of phone calls, are linked to subscription rates.
- H5 : Combinations of customer attributes along with history of the campaign can result in identification of those segments which have high subscription rates.

These hypotheses are exploratory in nature rather than causal. The EDA checks if the data support any of them and provides any patterns for further analysis.

Dataset Overview & Data Pre-Processing

1. Dataset Overview

1. Dataset Structure and Variables

Initially, the data consisted of 41,180 observations, each with 21 attributes regarding the customers who were contacted for a marketing campaign in the telecommunication sector. Attributes include demographic information, financial or debt-related information, past campaign information, current campaign information, and macroeconomic information, where y is the binary variable determining whether the client subscribed to the campaign or not (yes/no).

Variable group	**Variables**	**Type**
Demographic	*age, job, marital, education*	Mixed
Financial/debt	*default, housing, loan*	Categorical
Current campaign	*contact, month, day_of_week, duration, campaign*	Mixed
Previous campaign	*pdays, previous, poutcome*	Mixed
Macroeconomic	*emp.var.rate, cons.price.idx, cons.conf.idx, euribor3m, nr-employed*	Numerical
Target	*y*	Binary

2. Data Quality Assessment and Pre-Processing

The initial dataset comprised 41,180 records with 21 variables. After eliminating 12 duplicate records and ensuring that all the variable types were validated, the final analytical dataset comprised 41,168 records without missing values and duplicates. The target variable was imbalanced as it consisted of 11.26% of customers subscribing to it, thus acting as a benchmark for future comparisons.

Unknown categories were maintained when they comprised valid data recorded. The value of 999 in pdays was considered as a sentinel value for customers who had never been contacted before instead of the number of days, as it was present in 96.3% of records. The extreme values in the variables age, duration, and campaign were also kept as they could be valid in the context of the campaign.

Issue identified	Evidence	Treatment	Justification
CSV parsing	Dataset using ; delimiter	Imported using correct delimiter	Ensures that each variable is correctly separated
Data types	Numerical variables initially required type validation	Numerical variables retained in appropriate numeric formats	Enables valid numerical analysis
Documentation discrepancies	*balance* listed in data dictionary but absent in the dataset	Excluded from analysis	Cannot be reconstructed
Duplicates	12 duplicate observations identified	Removed	Prevents repeated records from influencing analysis
Missing values	No explicit NaN values identified	None	No missing-value imputation required
Unknown categories	Present in several categorical variables	Retained as explicit categories	No evidence justified imputation or removal
pdays = 999	39,655 observations (96%)	Retained as a sentinel value	Not interpreted as actual 999 days
Extreme values	Strong right tails in *duration* and *campaign*. High ages observed	Retained	Values are possible and not necessarily erroneous
age = 17	5 observations	Retained	Observations are coherent and age-majority rule is provided

No missing values (NaN) were explicitly found in the dataset. Some categorical columns had the category unknown in them; notably, the variables *default*, *housing*, *loan*, and *education*. The fact that unknown is not necessarily a missing value and does not require imputation was confirmed during data quality control.

The connection between unknowns and the dependent variable was assessed as well; for instance, there were 990 cases where both *housing* = unknown and *loan* = unknown, with a subscription rate close to the general one. As such, these rows were kept.

The variable *pdays* needs special consideration owing to the wide use of 999 as a sentinel value. There were 39,655 observations (96.3%) that have 999, whereas 1,513 observations had values from 0 to 27 days. Therefore, 999 was not considered as an actual number of days, and the observations were kept. Moreover, the association between *pdays*, *previous*, and *poutcome* was reviewed for logical consistency. It appeared that there was coherence between these variables because when previous was 0, then *poutcome* was nonexistent, and vice versa.

The numerical range test showed no negative values for the variables *duration*, *campaign*, *previous*, or *pdays*, as well as no ages greater than 100 years. There were five cases with *age* = 17, which were kept since these data are consistent and there was no information about the minimum age of clients.

Outliers were discovered in relation to age, duration, and campaigns. The outliers were maintained since there were no violations of logic and range and were realistic in relation to the campaigns. These observations were seen as consistent in the context of the campaign, not erroneous observations. For example, even though 4,918 is extremely high for call duration and 56 for campaign contacts, it could happen within the framework of the campaign. So, no observations were deleted due to their extremeness.

3. Final Analytical Dataset

Following data cleaning and validation, the final analytical dataset contains 41,168 observations and 21 variables. The 12 exact duplicate records identified during the initial audit were removed. No explicit missing values or duplicate observations remain, and the numerical and categorical variables are stored using appropriate data types.

Final dataset characteristic	Result
Observations	41,168
Variables	21
Missing values	0
Exact duplicated	0

The target variable is imbalanced, with 88.74% of observations corresponding to non-subscription and 11.26% to subscription. The overall subscription rate of 11.26% is therefore established as the baseline for the subsequent EDA. Segment-level subscription rates will be compared against this baseline to identify groups with comparatively higher or lower campaign response.

Exploratory Data Analysis

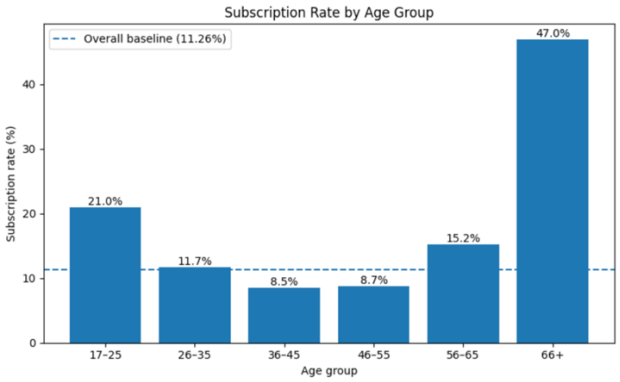
1. EDA Approach

The subscription rates were compared between categorical variables with the help of descriptive statistics and graphical representations. The chi-square and Cramer's V tests were used to analyze the relationships between categorical variables, while Pearson correlation was utilized for finding out the relations between numerical predictors. The cross-tabulation analysis was also performed to look for customer segments and dependencies.

Since the response variable is categorical, the EDA is also done in order to assist in the construction of a logistic regression model by providing insight on possible predictors and their interaction effects on each other.

2. Demographic Characteristics

The variable for age of the customer showed a non-linear pattern with respect to the subscription. For the group between 17-25 years old, the subscription ratio was 20.96% and for the group of 66 years and above, it was 47.00%. On the other hand, for the groups of 26-35, 36-45 and 46-55 years old, the ratios were 11.72%.

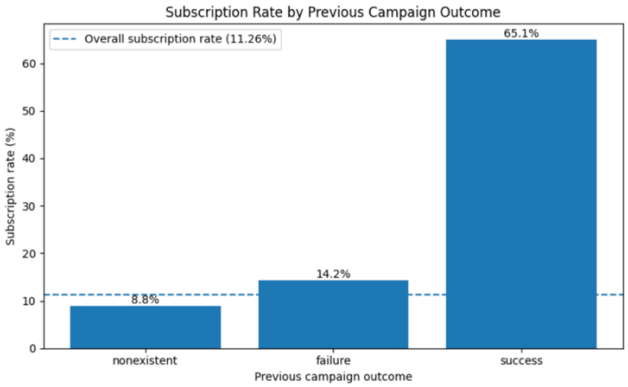


The dashed line represents the overall subscription rate of 11.26%

Both age group and occupation had a significant association with subscription ($p<0.001$), having a Cramér's V value of 0.167 and 0.153. The highest recorded subscription was recorded for students at 31.43%, followed by retired people at 25.23%, and blue-collar workers at 6.90%. However, these findings have a strong correlation with age as 51.9% of the students fell into the 17-25 category, whereas 79% of retired individuals were older than 56 years.

3. Previous Campaign History

The variable that had the highest relationship with subscription amongst the categorical variables was the outcome of the last campaign ($\chi^2 = 4{,}217.66$, $p < 0.001$; Cramér's V = 0.320). Customers who had a successful outcome from their last campaign had a 65.06% subscription rate, while those with unsuccessful campaigns had 14.24%, and those without outcomes had 8.83%.



The dashed line represents the overall subscription rate of 11.26%

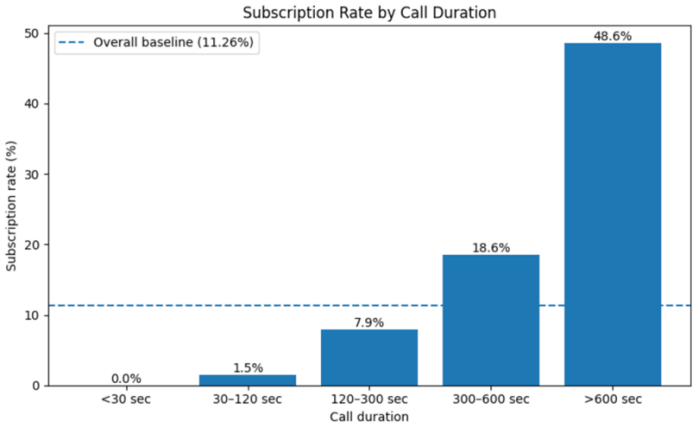
Previous contacts were also a determinant of subscription, but it seems that it is very closely linked with previous campaign outcome. Subscribers who had a previous campaign succeed on their part exhibited a very high rate of subscription even if there was only one previous contact (59.61%). In general, it seems that previous campaign success is more telling than previous contacts.

4. Current Campaign Characteristics

Call duration

Call duration displayed one of the clearest descriptive patterns in the dataset. Subscription rates increased consistently with longer calls, from 0.00% for calls shorter than 30 seconds to 48.61% for calls longer than 10 minutes.

duration	n	subscription_rate
<30 sec	1724	0.00
30–120 sec	11188	1.48
120–300 sec	17056	7.92
300–600 sec	7736	18.58
>600 sec	3464	48.61

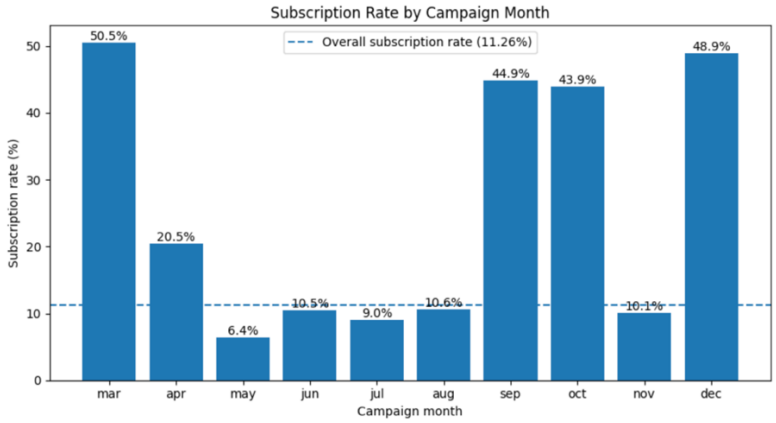


The dashed line represents the overall subscription rate of 11.26%

The call length emerged as one of the most powerful descriptive relationships with subscription. The rate of subscription was zero percent for calls under 30 seconds but climbed to 48.61 percent for calls over 10 minutes ($\chi^2 = 6{,}729.61$, $p < 0.001$; Cramér's V = 0.404). However, call length is measured after the customer's interaction has started and therefore can be caused by customer involvement rather than be its cause. It is not suitable for the pre-contact targeting model.

Campaign Timing and Contact Method

Subscription rates also varied substantially across campaign months. March had the highest observed rate (50.55%), followed by December (48.90%), September (44.91%) and October (43.93%), while May had the lowest rate at 6.44%.



The dashed line represents the overall subscription rate of 11.26%

Monthly subscription rates fluctuated considerably, with March recording the highest percentage at 50.55% and May the lowest at 6.44%. Month was significantly correlated with subscription ($p < 0.001$; Cramér's $V = 0.275$) and the second most significant categorical variable after previous campaign performance. However, the considerable variance in the number of samples within each month makes it difficult to consider this seasonal influence.

Subscription is also related to contact method. While this relationship was quite weak (Cramér's $V = 0.145$), there was still a higher rate of subscriptions with cellular contacts (14.73%) compared to that of telephone contacts (5.23%).

Customer Segmentation

In the customer segmentation analysis using age and previous campaign success, it was evident that there was large variation in the results. For the customers who had a previous campaign success, there were high levels of subscriptions within those who are 66 years old and above (76.5%) and between 56-65 years old (71.6%). On the other hand, those who had no previous campaign success exhibited low levels of response. This is evidence to support H_5.

Other Variables and Overall EDA Findings

Education, housing, loan, weekday, and campaign variables exhibited a weak or limited association to the subscription variable. The subscription levels decreased as the number of contacts rose. However, this might have been due to the law of diminishing returns or variability in customer responses. Similarly, the macroeconomic variables also demonstrated a weak correlation to the subscription, although they had strong correlations among themselves, especially emp.var.rate and euribor3m ($r = 0.972$).

In general, previous campaign outcome, campaign month, age, occupation, contact type, and call duration were found to have the strongest correlations with subscription. The results partially support H_1 since the differences among demographics are evident only in age and occupation; however, the results provide limited support for H_2 because of the weak relationship between financial and debt factors. H_3 is strongly supported by the correlation between previous campaign outcome and subscription; on the other hand, H_4 is partially supported as some campaign attributes have significant correlations. H_5 is also supported since high response rates were noticed in some age x previous campaign outcome combinations.

Overall, the EDA provides information about the modelling phase through discovering potential predictors, interactions, and dependencies among variables. Considering that there is a binary outcome, it becomes reasonable to use logistic regression. The dependence between age and campaign success in previous months suggests the analysis of interaction between these factors. High correlation between macroeconomic variables reveals the existence of the problem of multicollinearity which should be considered when modelling. It is necessary to handle call duration independently since it can be observed only during customer contact.